ModelIt!

Teacher's Guide

Designing a Cancer Drug That Actually Works

Multi-Scale Pharmacokinetic Modeling from Molecule to Patient

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-4, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS1-4, HS-ETS1-2

Use a model to illustrate the role of cellular division in producing and maintaining complex organisms; design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-scale computational model that connects molecular pharmacokinetics to cellular tumor response to organism-level treatment outcomes.
Disciplinary Core Idea	**LS1.A Structure and Function / ETS1.B Developing Possible Solutions** — Cellular processes including drug metabolism operate at multiple biological scales; engineering design requires systematic optimization of competing constraints.
Crosscutting Concept	**Systems and System Models** — Students model the human body as a complex system where drug behavior at the molecular scale produces emergent outcomes at the organism scale through multiple interacting subsystems.

CAST Testing Alignment

MULTIPLE CHOICE:

A cancer patient's blood work shows their liver metabolism rate is 40% faster than average. Based on your pharmacokinetic model, what is the most likely clinical consequence?

CONSTRUCTED RESPONSE:

Using your multi-scale model, explain why a drug that kills 99% of cancer cells in a petri dish might fail completely in a human patient. Reference at least three model components in your explanation.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to multi-scale pharmacokinetic modeling — one of the most powerful tools in modern drug development. Biotech skill focus: Multi-scale modeling (molecular to cellular to organism). The pharmaceutical industry loses billions of dollars annually on drugs that fail in clinical trials because researchers couldn't predict how molecular-level drug behavior would translate to patient-level outcomes. Computational pharmacokinetic models bridge this gap by integrating absorption, distribution, metabolism, and excretion (ADME) processes across biological scales.

Pharmacokinetics: The ADME Framework

Every drug's journey through the body follows four phases: Absorption (drug enters the bloodstream from the administration site — oral drugs must survive stomach acid and cross the intestinal wall, IV drugs bypass this entirely), Distribution (drug travels through the bloodstream and crosses into tissues — the blood-brain barrier, tumor vasculature, and protein binding all affect distribution), Metabolism (primarily in the liver, cytochrome P450 enzymes chemically modify the drug, usually inactivating it — genetic variation in CYP enzymes causes 10-fold differences in metabolism between patients), and Excretion (kidneys filter metabolites from blood into urine; some drugs are also excreted in bile, sweat, or exhaled air). The concentration-time curve that results from these four processes determines everything about whether a drug works or kills the patient.

The Therapeutic Window Problem in Oncology

Cancer drugs face a unique pharmacokinetic challenge: cancer cells are human cells. Unlike antibiotics, which target bacterial proteins absent in human cells, chemotherapy drugs must exploit the subtle differences between cancer cells and normal cells — usually faster division rates. This means the concentration required to kill cancer cells (minimum effective concentration, or MEC) is often dangerously close to the concentration that causes organ toxicity (maximum tolerated concentration, or MTC). The therapeutic window (MTC minus MEC) for drugs like doxorubicin, cisplatin, and paclitaxel can be as narrow as a 2-fold range, compared to 10-fold or wider for common antibiotics. This is why chemotherapy has such devastating side effects.

Multi-Scale Modeling in Drug Development

Traditional pharmacokinetic models operate at a single scale — tracking drug concentration in blood over time. Modern multi-scale models integrate three levels simultaneously: molecular scale (drug-receptor binding kinetics, enzyme-substrate interactions, protein conformational changes), cellular scale (signal transduction cascades, cell cycle arrest, apoptosis induction, resistance mechanism activation), and organism scale (organ function, immune response, clinical symptoms, survival probability). Physiologically-based pharmacokinetic (PBPK) models represent each organ as a compartment with its own blood flow, volume, and metabolic capacity. These models can predict drug behavior in populations that were never tested — pediatric patients, patients with liver disease, or patients taking multiple drugs simultaneously.

Metronomic Chemotherapy: A Modeling Success Story

Computational pharmacokinetic modeling led to a paradigm shift in cancer treatment called metronomic chemotherapy. Traditional protocols give the maximum tolerated dose (MTD) on an infrequent schedule (e.g., every 3 weeks) to allow the patient to recover from side effects between cycles. Models predicted that lower doses given more frequently could maintain drug levels within the therapeutic window more consistently, reducing the peaks that cause toxicity and the troughs where the drug drops below effective levels. Clinical trials confirmed this — metronomic dosing showed equal or better tumor control with dramatically fewer side effects for several cancer types. This is multi-scale modeling literally saving lives.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine pharmacokinetic components operating across molecular, cellular, and organism scales — from Drug Dosage input through absorption, blood concentration, tumor uptake, healthy cell damage, metabolism, clearance, therapeutic window, to side effect severity output.

E **Establish Relationships**

Students determine that Drug Dosage drives Blood Concentration through Absorption Rate, which simultaneously enables Tumor Uptake (desired) and Healthy Cell Damage (undesired), while Metabolism Rate and Kidney Clearance continuously remove the drug — creating the fundamental pharmacokinetic tension.

V **Visualize & Model**

Students build a computational model connecting all nine components across biological scales, visualizing the concentration-time curve and the therapeutic window where the drug is effective but not toxic.

E **Evaluate Outcomes**

Students run standard dosage, high-dose aggressive, and metronomic optimization scenarios to find dosing strategies that maximize time within the therapeutic window.

R **Revise & Extend**

Students add Tumor Blood Supply, Drug Resistance Factor, or Patient Liver Function to explore personalized pharmacokinetics and treatment optimization.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Over 95% of experimental cancer drugs fail in clinical trials. Not because they can't kill cancer cells — most of them can. They fail because we can't get them to kill cancer without killing the patient. Today, you're going to try to solve that problem."*

Watch For

Display a statistic about cancer drug failure rates. Let the magnitude sink in before continuing.

Slide 2: Learning Objectives

3 min

Say: *"You're going to build a multi-scale pharmacokinetic model — the same type of computational tool that pharmaceutical companies use to design drug dosing protocols. This is real biotech."*

Watch For

Pre-teach pharmacokinetics and therapeutic window. Have students write their own definitions before revealing the formal ones.

Slide 3: Big Question

3 min

Say: *"A drug kills 99% of cancer cells in a petri dish. Six years and two billion dollars later, it fails in human trials. What went wrong between the dish and the patient?"*

Watch For

Think-pair-share: Students brainstorm why a drug that works in a dish might fail in a body. Collect ideas on the board.

Slide 4: LEVER Framework

2 min

Say: *"We're going to connect molecular-level drug behavior to cellular-level tumor response to patient-level outcomes. That's multi-scale modeling — and it's how real drug companies design chemotherapy protocols."*

Watch For

Preview each LEVER step. Emphasize that this model spans three biological scales simultaneously.

Slide 5: Activity 1: Components

8 min

Say: *"These nine components represent the complete journey of a cancer drug through the body. Only ONE of them is truly under the doctor's control — which one?"*

Watch For

Guide identification of Drug Dosage as the only external component. Discuss why all other components are internal responses. Sort components by biological scale.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the cruel paradox of cancer treatment: the same pathway that delivers drug to the tumor also delivers it to every other organ in the body. How do you maximize one and minimize the other?"*

Watch For

Students trace the pathway from Drug Dosage through Blood Concentration to both Tumor Uptake and Healthy Cell Damage. Identify the therapeutic window as the narrow zone where the drug works without killing.

Slide 7: Activity 3: Simulation

10 min

Say: *"You're now the pharmacokineticist. Run these three scenarios and find the dosing strategy that keeps the drug inside the therapeutic window the longest."*

Watch For

Students predict outcomes first, then run standard, aggressive, and metronomic dosing scenarios. Track time-within-therapeutic-window as the key metric.

Slide 8: Discoveries

5 min

Say: *"You just discovered what took the pharmaceutical industry decades to figure out — sometimes less is more. Metronomic dosing, predicted by computational models, is now saving real cancer patients' lives."*

Watch For

Connect simulation findings to real clinical practice. Discuss how modeling predicted metronomic chemotherapy before clinical trials proved it.

Slide 9: STEM Challenge

12 min

Say: *"A pharmaceutical company just hired your team. Design an algorithm that adjusts chemotherapy doses in real time based on patient biomarkers. This is the future of cancer treatment."*

Watch For

Groups design adaptive dosing algorithms. Must include biomarker inputs, decision logic, and safety thresholds. Present and defend designs.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Drug Dosage
- **INTERNAL:** Absorption Rate, Blood Concentration, Tumor Uptake, Healthy Cell Damage, Metabolism Rate, Kidney Clearance, Therapeutic Window, Side Effect Severity

Relationships

- **Drug Dosage to Blood Concentration:** POSITIVE (+) — Higher drug dosage increases the amount of drug entering the bloodstream through absorption, raising blood concentration proportionally to the dose administered.
- **Blood Concentration to Tumor Uptake:** POSITIVE (+) — Higher blood concentration creates a steeper concentration gradient between blood and tumor tissue, driving more drug molecules across the tumor vasculature into cancer cells.
- **Blood Concentration to Healthy Cell Damage:** POSITIVE (+) — Higher blood concentration simultaneously exposes healthy tissues to more drug, increasing collateral damage to rapidly-dividing normal cells like bone marrow, gut lining, and hair follicles.
- **Metabolism Rate to Blood Concentration:** NEGATIVE (-) — Faster liver metabolism breaks down more drug per unit time, actively reducing blood concentration and shortening the duration of therapeutic drug levels.
- **Kidney Clearance to Blood Concentration:** NEGATIVE (-) — More efficient kidney filtration removes drug metabolites and some active drug from circulation faster, accelerating the decline of blood concentration after dosing.
- **Healthy Cell Damage to Side Effect Severity:** POSITIVE (+) — Greater healthy cell damage directly increases the severity of clinical side effects — nausea from gut lining damage, immunosuppression from bone marrow damage, hair loss from follicle damage.

Key Discoveries

- Drug concentration in the blood follows a predictable curve — rising after administration, peaking, then declining as metabolism and clearance remove the drug from circulation
- The therapeutic window for most cancer drugs is extremely narrow — the dose needed to kill cancer cells is dangerously close to the dose that damages healthy tissue
- Metabolism rate and kidney clearance vary significantly between patients, meaning the same dose can be therapeutic in one person and toxic in another
- Metronomic dosing (lower, more frequent doses) can maintain drug levels within the therapeutic window more consistently than traditional high-dose protocols

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More drug always means more effective treatment"	Increasing drug dosage beyond the therapeutic window does NOT improve outcomes — it increases toxicity and side effects without proportionally increasing tumor kill. Cancer cells that survive the initial drug exposure are often the resistant ones, so higher doses kill more healthy cells while selecting for resistant cancer cells. The relationship between dose and effectiveness plateaus while the relationship between dose and toxicity continues to climb.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Cancer drugs specifically target cancer cells"	Most traditional chemotherapy drugs do NOT specifically target cancer cells. They target rapidly-dividing cells — which includes cancer but also bone marrow, gut lining, hair follicles, and immune cells. This is why chemotherapy causes hair loss, nausea, and immunosuppression. The drug cannot tell the difference between a dividing cancer cell and a dividing healthy cell.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Computer models can perfectly predict drug outcomes"	Pharmacokinetic models are powerful but imperfect. They simplify the staggering complexity of human biology into manageable mathematical equations. Real patients have genetic variation, comorbidities, microbiome differences, psychological stress responses, and countless other factors no model fully captures. Models are tools for better decision-making, not crystal balls.